Physicist / Engineer physicist BSc, year III / semester 1
"Optics problem solving"

RETAKE MIDTERM TEST 1-2

16-12-2025

1 Design of an intensity attenuation filter

Let us determine the thickness of a pure aluminum layer, vacuum sputtered on glass substrate, necessary to reduce the transmitted intensity of a plane wave of normal incidence to 0.1% of its input intensity! Do consider surface reflections taking place on air/metal and metal/glass boundaries! (We can disregard multiple reflections, substrate/air reflection and interference effects.) The complex refractive index of aluminum at 633 nm is: $n = 1.450$ $\kappa = 7.539$. The substrate is B270 glass, $n_s = 1.521$. What is the reflectance on the air/aluminum and aluminum/glass boundaries? To what fraction does intensity that enters the aluminum need to be decreased after passing through a thin layer of *z* thickness? What is the necessary layer thickness? **25 p**

$$T = \frac{I_{\text{out}}}{I_{\text{in}}} = 0.1\% \tag{1}$$

$$R_{\text{air-alum}} = \left|\frac{n_{\text{air}} - \tilde{n}}{n_{\text{air}} + \tilde{n}}\right|^2 = \left|\frac{1 - n + i\kappa}{1 + n - i\kappa}\right|^2 = \left|\frac{1 - 1.450 + i7.539}{1 + 1.450 - i7.539}\right|^2 = \frac{57.04}{62.84} = 90.77\% \tag{2}$$

$$R_{\text{air-alum}} = \left|\frac{\tilde{n} - n_s}{\tilde{n} + n_s}\right|^2 = \left|\frac{n - i\kappa - n_s}{n - i\kappa + n_s}\right|^2 = \left|\frac{1.450 - i7.539 - 1.521}{1.450 - i7.539 + 1.521}\right|^2 = \frac{56.84}{65.66} = 86.57\% \tag{3}$$

$$T = (1 - R_{\text{air-alum}}) \cdot \frac{I(z)}{I_0} \cdot (1 - R_{\text{alum-subs}}) = 0.1\% \quad \rightarrow \quad \frac{I(z)}{I_0} = 8.067\% \tag{4}$$

$$\frac{I(z)}{I_0} = e^{-2k_{\text{im}}z} \;\; ; \;\; k_{\text{im}} = k_0 \cdot \kappa = \frac{2\pi}{\lambda_0} \cdot \kappa = 7.483 \cdot 10^7 \text{ m}^{-1} \tag{5}$$

$$z = -\frac{1}{2k_{\text{im}}} \cdot \ln\left(\frac{I(z)}{I_0}\right) = 16.82 \text{ nm}. \tag{6}$$

2 Analysis of a telephoto objective with matrix optics

We are examining a telephoto objective made of two thin lenses placed in air, see Fig. 1. The focal length of the lenses are f_1 = 100 mm, and f_2 = −80 mm. What is the refractive power of the lenses (P_1, P_2)? What should the distance d_1 between the lenses be to get a resultant effective focal length of f' = 200 mm? Write up the transfer matrix of the complete system from the entry surface of the first lens to the exit surface of the second, and determine its parameters! Trace a single ray from the first lens to the second that is parallel to the optical axis in the object field, and its height on the first lens is y_1 = 10 mm. What is the ray height on the second lens (y_2)? What is the angle that this ray makes with the optical axis after the second lens (θ_2)? Relative to the second lens at which d_2 distance can the focal spot be found? Based on this, where is the image-field principal plane relative to the second lens (z')? **25 p**

$$P_1 = \frac{1}{f_1} = 0.01 \text{ mm}^{-1} \;\; ; \;\; P_2 = \frac{1}{f_2} = -0.0125 \text{ mm}^{-1} \tag{7}$$

$$P = P_1 + P_2 - P_1 P_2 \frac{d_1}{n_1} = \frac{1}{f'} \;\; ; \;\; n_1 = 1 \tag{8}$$

$$d_1 = \frac{P_1 + P_2 - f'^{-1}}{P_1 P_2} = 60 \text{ mm} \tag{9}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -P_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{d_1}{n_1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -P_1 & 1 \end{bmatrix} \tag{10}$$

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ -P_2 & 1 \end{bmatrix} \begin{bmatrix} 1 - P_1 \frac{d_1}{n_1} & \frac{d_1}{n_1} \\ -P_1 & 1 \end{bmatrix} \tag{11}$$

$$\mathbf{M} = \begin{bmatrix} 1 - P_1 \frac{d_1}{n_1} & \frac{d_1}{n_1} \\ -P_1 - P_2 + P_1 P_2 \frac{d_1}{n_1} & 1 - P_2 \frac{d_1}{n_1} \end{bmatrix} = \begin{bmatrix} 0.4 & 60\ \text{mm} \\ -0.005\ \text{mm}^{-1} & 1.75 \end{bmatrix} \tag{12}$$

$$\begin{bmatrix} y_1 \\ q_1 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix} \tag{13}$$

$$\begin{bmatrix} y_2 \\ q_2 \end{bmatrix} = \mathbf{M} \begin{bmatrix} y_2 \\ q_2 \end{bmatrix} = \begin{bmatrix} 4\ \text{mm} \\ -0.05 \end{bmatrix} \tag{14}$$

$$q_2 = n_2 \theta_2 \quad ; \quad \theta_2 = -0.05\ [\text{rad}] \tag{15}$$

$$d_2 = -\frac{y_2}{\operatorname{tg} \theta_2} \approx -\frac{y_2}{\theta_2} = 80\ \text{mm} \tag{16}$$

$$f' = d_2 - z' \quad \rightarrow \quad z' = d_2 - f' = -120\ \text{mm}. \tag{17}$$

3 Analysis of a Fabry-Pérot resonator with multiple-beam interference

We are investigating a laser diode of λ_0 = 650 nm central wavelength having a Fabry-Pérot resonator made of GaInP compound semiconductor, the refractive index of which is n = 3.599 (the imaginary part can be neglected), the surrounding medium is air. What is the length (d) of the rectangle-shaped resonator, if the distance between two neighboring resonator modes is $\Delta\lambda_0 = 110$ pm? In which resonator mode (m) is the laser operating? The surfaces are uncoated, they are reflective only because of the high refractive index. Calculate the reflectance at the semiconductor/air boundary (R) for normal incidence! What is the finesse value of the resonator? By what $\Delta\lambda_0$ value can the wavelength grow so that the Airy function describing the transmittance decreases exactly to its half (see resolving power)? **25 p**

$$\delta_m = m \cdot 2\pi = \frac{2\pi}{\lambda_0} n2d \quad \Rightarrow \quad m = \frac{n2d}{\lambda_0} \tag{18}$$

$$m - 1 = \frac{n2d}{\lambda_0 + \Delta\lambda_0} \tag{19}$$

$$\mathrm{m} - (\mathrm{m} - 1) = \frac{n2d}{\lambda_0} - \frac{n2d}{\lambda_0 + \Delta\lambda_0} \tag{20}$$

$$\frac{1}{\frac{1}{\lambda_0} - \frac{1}{\lambda_0 + \Delta\lambda_0}} = n2d \tag{21}$$

$$d = \frac{\lambda_0}{2n} \frac{1}{1 - \frac{\lambda_0}{\lambda_0 + \Delta\lambda_0}} = 0.5337\ \text{mm} \tag{22}$$

$$m = \frac{n2d}{\lambda_0} = 5910 \tag{23}$$

$$R = \left| \frac{1 - n}{1 + n} \right|^2 = 31.94\% \tag{24}$$

$$\mathcal{F} \triangleq \frac{\pi\sqrt{R}}{1-R} = 2.609 \tag{25}$$

$$\mathcal{R} \triangleq \left|\frac{\lambda_0}{\Delta\lambda_0}\right| = m \cdot \mathcal{F} = 15417 \quad \rightarrow \quad \Delta\lambda_0 = 42.17 \text{ pm}. \tag{26}$$

4 Analysis of the far-field diffraction of an aperture by Fraunhofer approximation

You are examining the far-field diffraction pattern of a Σ aperture presented in Fig. 2 with light of λ wavelength, the illumination is a plane wave of z-direction with E_0 = 1 V/m amplitude. Determine the complex field distribution on a plane screen placed at a large z' distance! How can we write up the far-field amplitude distribution of the rectangle on the right? Based on this, how can we write up that of the rectangle on the left? If we add up the two, what simple interpretation can be associated with the obtained field distribution? **25 p**

$$E(x',y') = \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}\left(x'^2+y'^2\right)}}{i \cdot \lambda z'} \iint_{\Sigma} E(x,y) \cdot e^{-i\frac{k}{z'}(x'x+y'y)} dxdy \tag{27}$$

For the rectangle on the right:

$$INT_{\mathrm{r}} \triangleq \iint_{\Sigma \mathrm{r}} e^{-i\frac{k}{z'}(x'x+y'y)} dxdy = \int_0^{2a} e^{-i\frac{k}{z'}x'x} dx \int_{-2a}^{4a} e^{-i\frac{k}{z'}y'y} dy \tag{28}$$

$$\int_0^{2a} e^{-i\frac{k}{z'}x'x} dx = \frac{z'}{-ikx'}\left[e^{-i\frac{k}{z'}x'x}\right]_0^{2a} = \frac{z'}{-ikx'}\left[e^{-i\frac{k}{z'}x'2a} - 1\right] = \tag{29}$$

$$= \frac{z'e^{-i\frac{k}{z'}x'a}}{-ikx'}\left[e^{-i\frac{k}{z'}x'a} - e^{i\frac{k}{z'}x'a}\right] = \frac{2z'e^{-i\frac{k}{z'}x'a}}{kx'}\sin\left(\frac{k}{z'}x'a\right) = 2a \cdot e^{-i\frac{k}{z'}x'a}\,\mathrm{sinc}\left(\frac{2ax'}{\lambda z'}\right) \tag{30}$$

$$\int_{-2a}^{4a} e^{-i\frac{k}{z'}y'y} dy = 6a \cdot e^{-i\frac{k}{z'}y'3a}\,\mathrm{sinc}\left(\frac{6ay'}{\lambda z'}\right) \tag{31}$$

$$INT_{\mathrm{r}} = 12a^2 e^{-i\frac{k}{z'}(x'a+y'3a)}\,\mathrm{sinc}\left(\frac{2ax'}{\lambda z'}\right)\mathrm{sinc}\left(\frac{6ay'}{\lambda z'}\right) \tag{32}$$

For the rectangle on the left:

$$INT_{\mathrm{l}} = 12a^2 e^{i\frac{k}{z'}(x'a+y'3a)}\,\mathrm{sinc}\left(\frac{2ax'}{\lambda z'}\right)\mathrm{sinc}\left(\frac{6ay'}{\lambda z'}\right) \tag{33}$$

$$INT_{\mathrm{r}} + INT_{\mathrm{l}} = 24a^2\,\mathrm{sinc}\left(\frac{2ax'}{\lambda z'}\right)\mathrm{sinc}\left(\frac{6ay'}{\lambda z'}\right)\cos(x'a + y'3a) \tag{34}$$

$$E(x',y') = \frac{e^{ikz'} \cdot e^{i\frac{k}{2z'}\left(x'^2+y'^2\right)}}{i \cdot \lambda z'} \cdot 24a^2\,\mathrm{sinc}\left(\frac{2ax'}{\lambda z'}\right)\mathrm{sinc}\left(\frac{6ay'}{\lambda z'}\right)\cos(x'a + y'3a)\,. \tag{35}$$

This is the field distribution of an aperture of $2a$×$6a$ size, multiplied by a two-beam interferogram having oblique fringes.

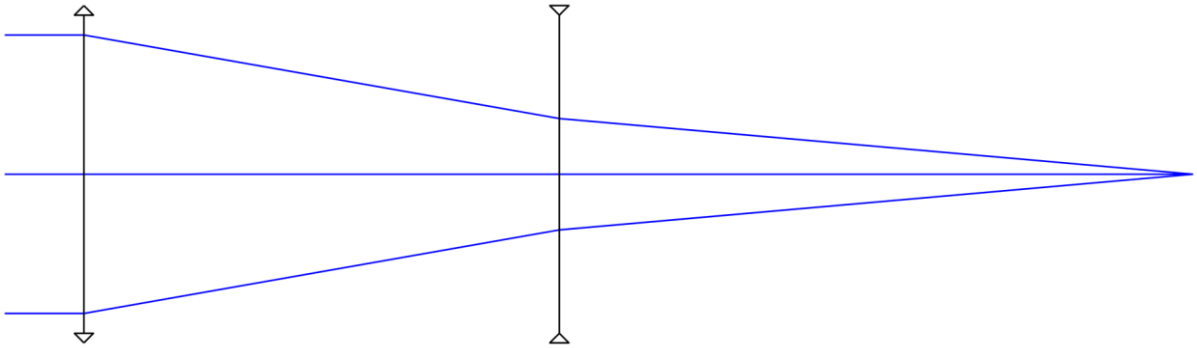


Fig. 1

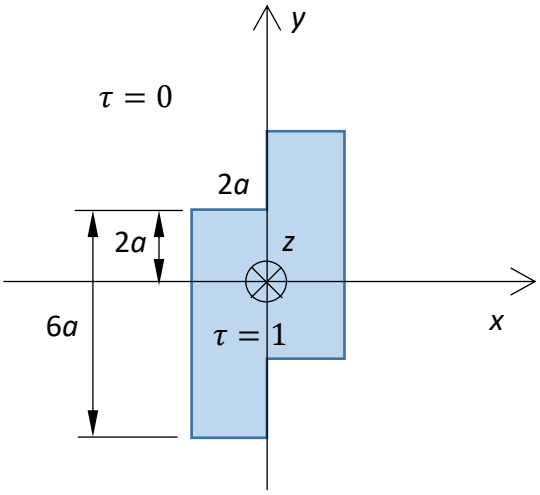


Fig. 2